



AWS Academy Cloud Foundations – Configuring Elastic Block Storage, Relational Databases, Load Balancing, and Autoscaling

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Lab 4 – Working with EBS

Purpose

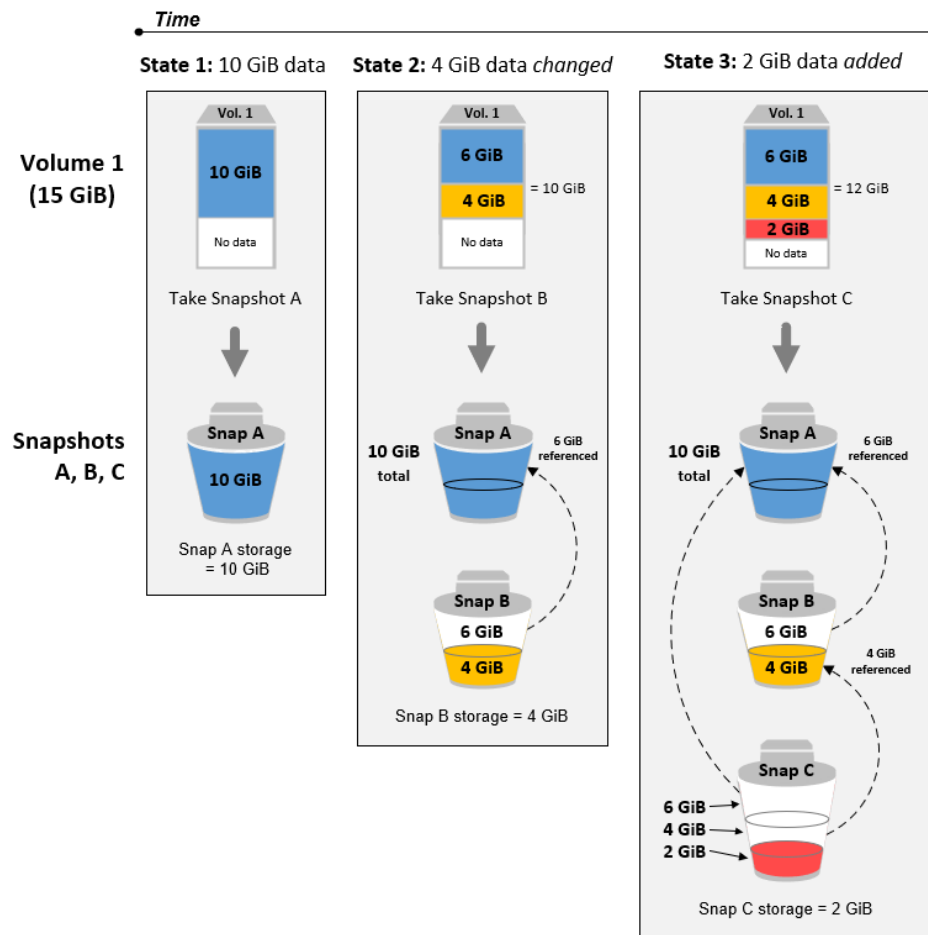
This lab helps to develop a deeper understanding of how to manage and optimize EC2 instances using the Elastic Block Store to save EC2 instance states. This lab also develops the skills required to effectively use EBS Snapshots and Volumes using the AWS Console to provide persisting memory, durability, and reliability. Because EBS allows EC2 instances to work just how it was before the instance stopped or failed, it proves a valuable skill to learn.

Background

EBS or Elastic Block Store is an AWS service to store information used in EC2 instances. EBS operates using Amazon EBS volumes and Amazon EBS snapshots; volumes are storage that are attached to EC2 instances like a hard drive attached to a computer. A Snapshot is a point in time backup of an EBS volume that exists independently of the volume itself; it can be used to save the state of an EC2 instance to provide backup and recovery. Block storage is the type of storage that EBS uses; block storage allows an object that is stored to be partially updated without needing to reupload the entire object under storage. This system allows for faster and minimal updates compared to object storage, which requires the entire object to be reuploaded after updating the entire object. When an EC2 Volume or Snapshot needs to be updated, only the blocks of the object that have been changed from the previous volume or Snapshot are reuploaded. For example, if an EBS snapshot had two blocks and an update is needed if only one of those blocks had a change in value, only that block is reuploaded to show the new values.

EBS is a vital part of any AWS solution that requires the use of EC2 instances. EBS provides redundancy, durability, and reliability that is needed for all AWS solutions. This allows for EC2 instance memory to stay persistent even when an EC2 instance unexpectedly shuts down or is manually stopped. This eliminates the period where the EC2 instance state needs to load into the RAM of the instance, allowing fast recovery time which may be necessary for high intensity applications and AWS solutions. To ensure durability of data within EBS volumes, they are replicated within their availability zone in the event of a hardware malfunction. To ensure that EBS volumes stay up to date, EBS Snapshots are taken to save the state of the EBS Volume. These snapshots can be monitored using CloudWatch Events and are stored using Amazon S3. Having an EBS volume allows for multiple EC2 instances to have identical configurations which provide consistency when delivering a solution to end users and businesses requiring the cloud solution. Volumes are changed constantly as they operate live rather than statically, having a snapshot of the EBS volume and then converting that snapshot into AMI or Amazon Machine Image allows for redundancy when EC2 instances fail, and allows for other EC2 instances to have updated identical EBS volumes when the memory of one EC2 instance is updates with information what all those instances may want. Another way of ensuring all EC2 instances to share the same data in an EBS volume is to use the EBS Multi-Attach feature which allows for one EBS volume to be attached to multiple instances.

Snapshots are critical to saving EBS volumes which allow for redundancy in AWS solutions. When a snapshot is first taken of an EBS volume, it takes a full snapshot, meaning that includes all the data blocks of the EBS volume at the time of creating the snapshot, essentially replicating the EBS volume. Any subsequent snapshots taken of the same EBS volumes are called an incremental snapshot, meaning only the data blocks of that volume that have changed will be uploaded into the S3 bucket containing the snapshots. This action does not replace the old data blocks but rather stores them separately.



docs.aws.amazon.com

This process is identical when taking a snapshot of a different volume as well. An example of this is if a snapshot of EBS volume A which is 10GB is taken and restored into a different EBS volume, once a snapshot of the new EBS volume is taken it will reference the original snapshot taken of volume A and store the new data from the new EBS volume separately. Understanding how EBS snapshots work allows for cloud practitioners to provide data durability, rapid disaster recovery, and optimize costs for storage through incremental backups.

Lab 5 – Build a Database Server

Purpose

This lab helps to develop an understanding of database servers, the different types, how they operate, and their application in AWS solutions. This lab helps to develop an understanding of how different types of servers operate and why some types of servers are better suited for some applications compared to others. Understanding how data servers work is essential to also understand other aspects of the IT field like API requests, data structures, and possibly most importantly cost optimization.

Background

Database servers are dedicated pieces of hardware to manage, store, and provide access to data. There are 3 different types of database structures, structured data, semi-structured data, and unstructured data. Structured data is data that is formatted in an organized way; all information within a structured database fits within a predefined schema, typically arranged in rows and columns. All data within a structured database server is easily searchable, actionable, and easily interpretable by machines using SQL or Structured Query Language. Semi-Structured data is less organized than structured data but can still be queried by SQL. Semi structured data is not set to a rigid schema like structured data but uses tags, markers, and metadata to create a flexible hierarchy. By tagging data in semi-structured databases, it allows for a level of organization that allows for data to be easily searchable and queried by SQL. JSON or JavaScript Object Notation is most used with semi-structured data because it takes advantage of the flexibility of semi-structured data and combines it with the queryability of structured systems. JSON itself is a text-based format used to transport data, rather than JSON being a part of semi-structured data. It is that data being stored in JSON format that makes the data type semi-structured. JSON allows for data within an object to constantly change and still be queryable by adding new meta data fields to the object itself, this cannot be done in structured data formats as the extra field cannot be added to define an object better due to the rigid schema. On top of this, because of JSON's flexibility and query ability, virtually all platforms support JSON allowing JSON to provide further utility. Finally, the final data type is unstructured data, around 80% to 90% for all data in the world is stored as unstructured data. Some common examples of unstructured data include text, videos, audio, and social media. Querying using unstructured data takes a different approach than semi-structured or structured data because instead of using meta data or columns, it typically uses machine learning to convert the data into high-dimensional vector embeddings to find data based on conceptual similarity to query rather than keywords. This can be slightly oversimplified by imagining a program assigns values to an object stored in unstructured space, similar objects will have similar values assigned to them while more differing objects will have more differing values, each object will have multiple of these values, then your query will be assigned values and the object with the closest values to your query will be pulled.

AWS has different services for each type of data. The service for structured data is Amazon RDS (Relational Database Service), semi-structured data: Amazon Redshift and DynamoDB; unstructured data: S3. Amazon also offers another service for unstructured data: Amazon OpenSearch Service for indexing and searching through

unstructured data. Amazon RDS is a fully managed service by Amazon, meaning all the AWS account holder must do is put data into the database. In any fully managed service by Amazon all the users must do is focus on creating the application by writing code or managing the data. Amazon RDS automatically optimizes your database for scaling, provisioning, and read/write operations. It offers pay by the hour cost structure or leased for one to three years as a reserved instance which can offer up to 70% cost reduction by the hour. Amazon Redshift is an Amazon service mainly for analyzing data at a petabyte large scale. Redshift uses RA3 instances to create clusters to process data, these cluster nodes are further broken down into slices to better provision processing power of a task across slices. Before tasks are given to RA3 instances they are delegated by a leader node, the leader node receives the SQL query, parses them and creates an optimized query execution plan to provision processing power across compute nodes. Redshift RA3 instances automatically offload data into S3 storage lakes, when trying to access the S3 data Redshift Spectrum is used to access the data without needing to use any RA3 instances. Redshift allows integrations with other Amazon databases like Aurora and DynamoDB, and offers a low cost, scalable, and secure petabyte-scale cloud data warehouse service, designed for high-performance analytics, BI, and SQL querying. DynamoDB is like redshift, it supports semi-structured data, but DynamoDB excels more for storing and providing access to information rather than analytics. DynamoDB uses key-value pairs along with object metadata to provide consistently fast, single-digit millisecond latency for reading and writing at any scale. It is also a fully managed service allowing AWS architects to only focus on managing data within DynamoDB.

Lab 6 – Scale and Load Balance your Architecture

Purpose

This lab teaches practical skills of how to effectively utilize EC2 instances to scale and provision computing resources in a practical environment. This lab develops understanding of how real applications handle volatile workloads to optimize cost while still maintaining computing power to provide an effective AWS cloud solution. Having the skill to scale and load balance any workload is an essential skill to have and understand in virtually all positions in the IT field, it allows for better integration, and cost optimization when providing a service or product that has a non-constant level of computing resources need to provide low-latency solutions.

Background

Scaling and load balancing architecture are a key advantage that cloud computing has over traditional networks. Businesses can easily scale computing resources horizontally by both decreasing and increasing computing instances to optimize cost while still maintaining system functionality. In traditional networks businesses need to estimate necessary computing and storage capacity over the course of several years due to the high upfront costs associated with purchasing and maintaining hardware on-site. This lab only covers scaling and loads balancing EC2 instances. In AWS, resources like containers and database resources can also be scaled up and down to fit the capacity needed for the workload as well.

Autoscaling can either be configured more manually with autoscaling groups or through a managed service like AWS Elastic Beanstalk and AWS Fargate. AWS Elastic Beanstalk is a Platform as a Service for autoscaling EC2 instances. Fargate is similar but for containers instead. The Cloud Foundations lab uses autoscaling groups to create elastic architecture. These Autoscaling groups can be configured to create and dispose of new EC2 instances when certain usage percentages are hit. This also applies to another scalable resources like containers and database instances. Although the quantity of instances cannot increase with autoscaling the number of Read Writes is scaled instead to fit operating capacity. Autoscaling groups use metrics obtained by CloudWatch to trigger the creation or termination of EC2 instances to meet the varying computing needs of an AWS solution. This means CloudWatch metrics needs to be enabled on the AWS account for any autoscaling service like Fargate and Elastic Beanstalk to work. Once the metrics are enabled CloudWatch will provide the EC2 instance load as a percentage so that the autoscaling services can work as configured.

Load balancing is a crucial part for autoscaling and efficiency in any cloud and traditional networking architecture. Load balancing is a practice of distributing workload across multiple nodes to process the workload. Effective load balancing allows for those nodes to operate at capacity that lies between overusing and under using the available work capacity. This concept can be applied outside of the tech field, just like if a sales department has a monthly quota that quota is split between the employees so that that goal can be reached without overworking a small number of employees or wasting money on underused hires. The mechanism to split the workload created by queries in a network is called a load balancer. Load balancers check the health of computing nodes and does not send traffic to unhealthy nodes, unhealthy nodes are simply instances that cannot process traffic due to an error or it has reached capacity. This lets computing nodes effectively compute traffic so that AWS solutions can function properly. Load balancers in AWS also need to be configured with a listener which defines what traffic the load balancer is listening for, this way EC2 instances can be optimized for handling specific traffic if needed rather than general traffic generated by the AWS solution.

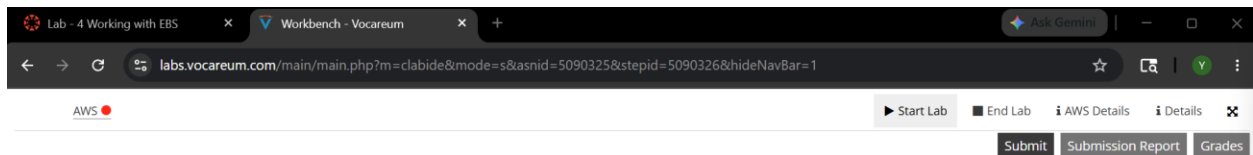
Lab 4 Walkthrough– Working with EBS

Lab Summary

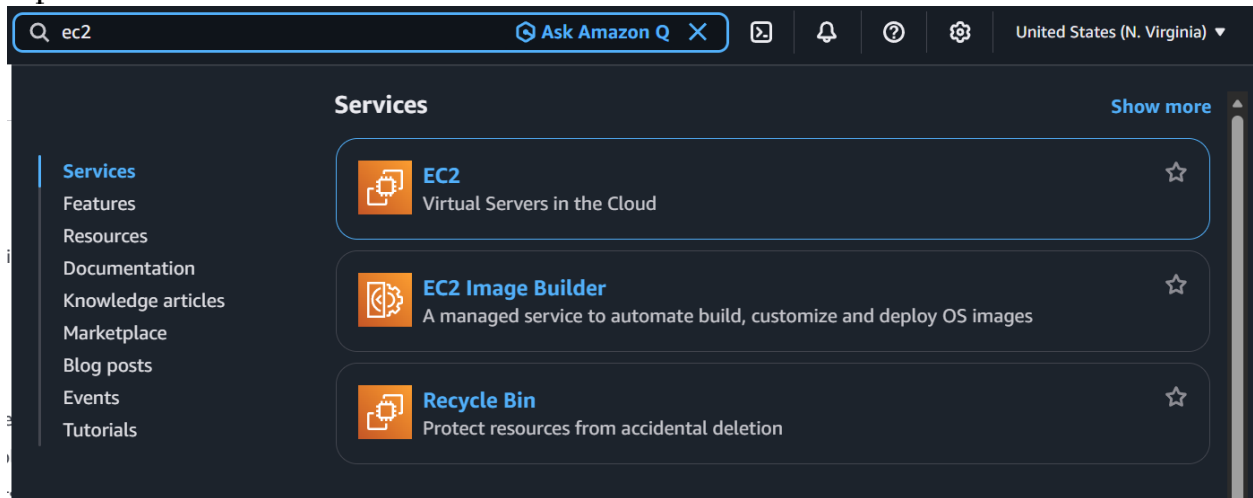
This lab is done through AWS Academy’s Cloud foundations course and walks through the steps of creating an EBS Volume, attaching it to an EC2 instance through a terminal interface, and creating snapshot backups of active EBS volumes. The EC2 instance is already created to allow the user to focus on the EBS aspect of the lab.

Lab Commands

Click the “Start Lab” button at the top right corner of the screen to start the lab and click the AWS with the green dot to open it.



Once the AWS management console is opened use the search bar located in the top left or find in the services menu.



Then, in the left navigation pane choose “Volumes”

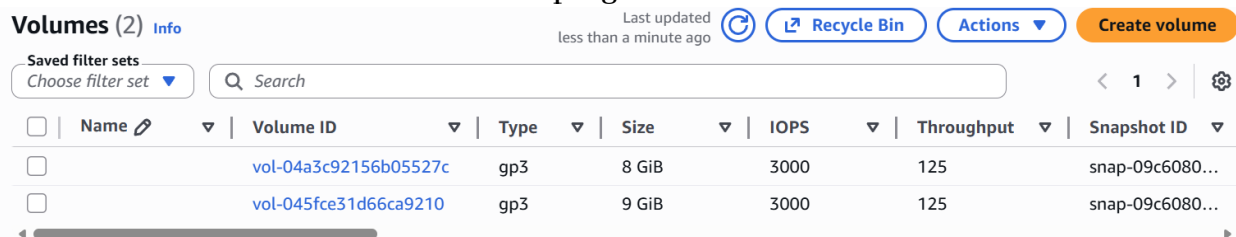
▼ Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Click “Create Volume” on the top right of the menu



Configure the EBS Volume for the following:

Type: General Purpose SSD (gp2, use this for the lab even if there is a newer version)

Size (GiB) 1

Availability zone: same as the “lab” EC2 instance

Tag: Key: Name | Volume: My Volume

Create volume [Info](#)

Create an Amazon EBS volume to attach to any EC2 instance in the same Availability Zone.

Volume settings

Volume type [Info](#)

General Purpose SSD (gp2)

Size (GiB) [Info](#)

1

Min: 1 GiB, Max: 16384 GiB.

IOPS [Info](#)

100 / 3000

Baseline of 3 IOPS per GiB with a minimum of 100 IOPS, burstable to 3000 IOPS.

Throughput (MiB/s) [Info](#)

Not applicable

Availability Zone [Info](#)

use1-az1 (us-east-1a)

Snapshot ID - optional [Info](#)

Don't create volume from a snapshot

Encryption [Info](#)

Use Amazon EBS encryption as an encryption solution for your EBS resources associated with your EC2 instances.

☐ Encrypt this volume

Tags - optional [Info](#)

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name

Value - optional

Q My Volume

Remove

Add new tag

You can add up to 49 more tags.

Snapshot summary [Info](#)

Click refresh to view backup information

The volume type that you select and the tags that you assign determine whether the volume will be backed up by any Data Lifecycle Manager policies.

Cancel

Create volume

Create the volume.

Next, select the newly created volume and attach it to the “Lab” Instance

Volumes (1/3) [Info](#) Last updated less than a minute ago [Recycle Bin](#) [Actions](#) [Create volume](#)

Saved filter sets Choose filter set [Search](#)

	Name	Volume ID	Type	Size
☒	My Volume	vol-08efa04dc9572d3d6	gp2	1 GiB
☐		vol-04a3c92156b05527c	gp3	8 GiB
☐		vol-045fce31d66ca9210	gp3	9 GiB

- Modify volume
- Create snapshot
- Create snapshot lifecycle policy
- Delete volume
- Attach volume
- Detach volume
- Force detach volume
- Manage auto-enabled I/O
- Copy volume - new
- Manage tags
- Resilience testing
Previously fault injection

< 1 > [Settings](#)

Snapshot ID
-
snap-09c6080...
snap-09c6080...

Attach volume [Info](#)

Attach a volume to an instance to use it as you would a regular physical hard disk drive.

Basic details

Volume ID
 vol-08efa04dc9572d3d6 (My Volume)

Availability Zone
use1-az1 (us-east-1a)

Instance [Info](#)

i-0299619ab1a696007
(Lab) (running)



Only instances in the same Availability Zone as the selected volume are displayed.

Device name [Info](#)

/dev/sdf

Recommended device names for Linux: /dev/xvda for root volume. /dev/sd[f-p] for data volumes.

 Newer Linux kernels may rename your devices to /dev/xvdf through /dev/xvdp internally, even when the device name entered here (and shown in the details) is /dev/sdf through /dev/sdp. 

[Cancel](#)

[Attach volume](#)

Next, connect to the EC2 Instance. Navigate to the EC2 Instances using the left navigation panel and select the “Lab” instance and choose connect right above it.

Instances (1/2) [Info](#)  [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#) 

[All states](#)  < 1 > 

	Name 	Instance ID	Instance state 	Instance type 	Status check	Alarm status
☐	Bastion Host	i-078bc47f2a2919950	 Running  	t2.micro	 2/2 checks passed	View alarms 
☒	Lab	i-0299619ab1a696007	 Running  	t2.micro	 2/2 checks passed	View alarms 

Choose the session Manager tab then choose connect.

Connect [Info](#)

Connect to an instance using the browser-based client.

Instance ID
 i-0299619ab1a696007 (Lab)

VPC ID
 vpc-080a27e1a6e94fa8e

Security groups
 sg-03064918b66cb7005
(c198898a5090326113704952t1w5862623
09869-Ec2SecurityGroup-qhQDhnf84PUQ)

IAM role
 LabRole

EC2 Instance Connect

SSM Session Manager

SSH client

EC2 serial console

Instance ID
 i-0299619ab1a696007 (Lab)

Connection type

☒ **Connect using a Public IP**
Connect using a public IPv4 or IPv6 address

☐ **Connect using a Private IP**
Connect using a private IP address and a VPC endpoint

☒ **Public IPv4 address**
 13.220.170.208

☐ **IPv6 address**
—

Username
Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ec2-user.



 **Note:** In most cases, the default username, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

[Cancel](#)

[Connect](#)

user

Next, view the storage available on the instance using this command: **df -h**

```

#_
~\_####_      Amazon Linux 2023
~~~\_#####\
~~~\_####|\
~~~\_#\|
~~~\_#/
~~~\_V~'_'->
~~~~
~~~~
~~~.~.~
~~~\_~\_~
~~~\_~\_~
~~~\_m/'_'~
[ec2-user@ip-10-1-11-84 ~]$ df -h

```

Filesystem	Size	Used	Avail	Use%	Mounted on
devtmpfs	4.0M	0	4.0M	0%	/dev
tmpfs	481M	0	481M	0%	/dev/shm
tmpfs	193M	448K	192M	1%	/run
/dev/xvda1	8.0G	1.6G	6.4G	20%	/
tmpfs	481M	0	481M	0%	/tmp
/dev/xvda128	10M	1.3M	8.7M	13%	/boot/efi
tmpfs	97M	0	97M	0%	/run/user/1000

Notice how the original 8GB disk volume is shown but the new one is not.

Create an ext3 file system on the new volume:

Sudo mkfs -t ext3 /dev/sdb

Create a directory for mounting the new volume:

Sudo mkdir /mnt/data-store

Mount the new volume

Sudo mount /dev/sdb /mnt/data-store

To configure the linux instance to mount the volume whenever the instance is started use the following command:

```
Echo "/dev/sdb /mnt/data-store ext defaults,noatime 1 2" | sudo tee -a /etc/fstab
```

Now view the storage available again:

```
[ec2-user@ip-10-1-11-84 ~]$ df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
devtmpfs	4.0M	0	4.0M	0%	/dev
tmpfs	481M	0	481M	0%	/dev/shm
tmpfs	193M	448K	192M	1%	/run
/dev/xvda1	8.0G	1.6G	6.4G	20%	/
tmpfs	481M	0	481M	0%	/tmp
/dev/xvda128	10M	1.3M	8.7M	13%	/boot/efi
tmpfs	97M	0	97M	0%	/run/user/1000
/dev/xvdf	975M	60K	924M	1%	/mnt/data-store

```
[ec2-user@ip-10-1-11-84 ~]$
```

Notice the extra 924Mb the volume provided.

Create a text file on the newly mounted volume using this command:

```
sudo sh -c "echo some text has been written > /mnt/data-store/file.txt"
```

Now create an Amazon EBS snapshot. Navigate back to the management console and reselect the created volume. And use the drop down menu on the top right to create a new snapshot.

Volumes (1/3) Info Last updated 9 minutes ago [Recycle Bin](#) [Actions](#) [Create volume](#)

Saved filter sets [Choose filter set](#)

	Name	Volume ID	Type	Size
☒	My Volume	vol-08efa04dc9572d3d6	gp2	1 GiB
☐		vol-04a3c92156b05527c	gp3	8 GiB
☐		vol-045fce31d66ca9210	gp3	9 GiB

- Modify volume
- Create snapshot**
- Create snapshot lifecycle policy
- Delete volume
- Attach volume
- Detach volume
- Force detach volume
- Manage auto-enabled I/O
- Copy volume - new
- Manage tags
- Resilience testing
 - Previously fault injection

Snapshot ID

Snapshot ID
-
snap-09c6080...
snap-09c6080...

Add a tag

Key: Name | Value: My snapshot

Tags Info A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key Value - optional [Remove](#)

[Add new tag](#)

You can add up to 49 more tags.

[Cancel](#) [Create snapshot](#)

In the left navigation pane choose snapshots and notice the newly created snapshot.

Snapshots (1) Info Last updated less than a minute ago [Recycle Bin](#) [Actions](#) [Create snapshot](#)

Owned by me

	Name	Snapshot ID	Full snapshot size	Volume size	Description	Storage
☐	My Snapshot	snap-0999bae2680b04205	53 MiB	1 GiB	-	Stand...

Now navigate back to the terminal session connectin to the “Lab” EC2 instance and delete the file created on the volume using this command:

```
Sudo rm /mnt/data-store/file.txt
```

Verify that it has been deleted using this command:

```
ls /mnt/data-store
```

now restore the text file using the snapshot. In the AWS Management Console select “My Snapshot”. Then choose the create volume from snapshot from the drop down menu.

Snapshots (1/1) [Info](#)

Last updated 1 minute ago

[Recycle Bin](#) [Actions](#) [Create snapshot](#)

Owned by me

☒	Name	Snapshot ID	Full snapshot size
☒	My Snapshot	snap-0999bae2680b04205	53 MiB

Create volume from snapshot

Create image from snapshot

Copy snapshot

Launch copy duration calculator

Delete snapshot

Manage tags

Snapshot settings

Archiving

< 1 >

Storage

Standards

Choose the same availiblilty zone and add a tag:
Key: Name | Value: Restored Volume
Create the volume.

Availability Zone [Info](#)

use1-az1 (us-east-1a)

Fast snapshot restore [Info](#)

Not enabled for selected snapshot

Volume initialization rate (MiB/s) - new, optional [Info](#)

Specify the rate at which the snapshot blocks are to be downloaded from Amazon S3 to the volume. [Additional costs apply](#)

Min: 100 MiB/s, Max: 300 MiB/s.

Encryption [Info](#)

Use Amazon EBS encryption as an encryption solution for your EBS resources associated with your EC2 instances.

☐ Encrypt this volume

Tags - optional [Info](#)

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Value - optional

[Remove](#)

[Add new tag](#)

You can add up to 49 more tags.

Snapshot summary [Info](#)

Click refresh to view backup information

The volume type that you select and the tags that you assign determine whether the volume will be backed up by any Data Lifecycle Manager policies.

[Cancel](#)

[Create volume](#)

Create volume [Info](#)

Create an Amazon EBS volume to attach to any EC2 instance in the same Availability Zone.

Volume settings

Snapshot ID

snap-0999bae2680b04205 (My Snapshot)

Volume type [Info](#)

General Purpose SSD (gp3) ▼

Size (GiB) [Info](#)

1

Min: 1 GiB, Max: 65536 GiB.

IOPS [Info](#)

3000

Min: 3000 IOPS, Max: 80000 IOPS.

Throughput (MiB/s) [Info](#)

125

Min: 125 MiB, Max: 2000 MiB, Baseline: 125 MiB/s.

Now navigate to volumes and attach the restored volume to the EC2 instance and choose `/dev/sdc/` or any next available identifier from the dropdown menu.

Volumes (1/4) [Info](#)

Last updated
less than a minute ago

Recycle Bin

Actions ▲

Create volume

Saved filter sets

Choose filter set ▼

Search

	Name	Volume ID	Type	Size
☐	My Volume	vol-08efa04dc9572d3d6	gp2	1 GiB
☐		vol-04a3c92156b05527c	gp3	8 GiB
☒	Restored Volu...	vol-0c82705d12cf7aa91	gp3	1 GiB
☐		vol-045fce31d66ca9210	gp3	9 GiB

- Modify volume
- Create snapshot
- Create snapshot lifecycle policy
- Delete volume
- Attach volume
- Detach volume
- Force detach volume
- Manage auto-enabled I/O
- Copy volume - *new*
- Manage tags
- Resilience testing
 - Previously fault injection

< 1 >
Snapshot ID ▼
-
snap-09c6080...
snap-0999bae...
snap-09c6080...

Attach volume [Info](#)

Attach a volume to an instance to use it as you would a regular physical hard disk drive.

Basic details

Volume ID

vol-0c82705d12cf7aa91 (Restored Volume)

Availability Zone

use1-az1 (us-east-1a)

Instance [Info](#)

i-0299619ab1a696007
(Lab) (running)



Only instances in the same Availability Zone as the selected volume are displayed.

Device name [Info](#)

/dev/sdg ▼

Recommended device names for Linux: `/dev/xvda` for root volume, `/dev/sd[f-p]` for data volumes.

Newer Linux kernels may rename your devices to `/dev/xvdf` through `/dev/xvdp` internally, even when the device name entered here (and shown in the details) is `/dev/sdf` through `/dev/sdp`.

Cancel

Attach volume

Next, Navigate back to the terminal session and create a new directory for mounting the new volume:

```
sudo mkdir /mnt/data-store2
```

Mount the new volume:

```
Sudo mount /dev/sdc /mnt/data-store2
```

Verify that volume you mounted has the file that you created earlier.

```
Ls /mnt/data-store2/
```

You can now see file.txt again

Submit your work and get full score:

Total score	25/25
Task 1 - Create EBS volume	5/5
Task 2 - Attach volume	5/5
Task 4 - Volume mounted	5/5
Task 5 - Snapshot created	5/5
Task 6 - Snapshot restored	5/5

Lab 5 Walkthrough– Build a Database Server

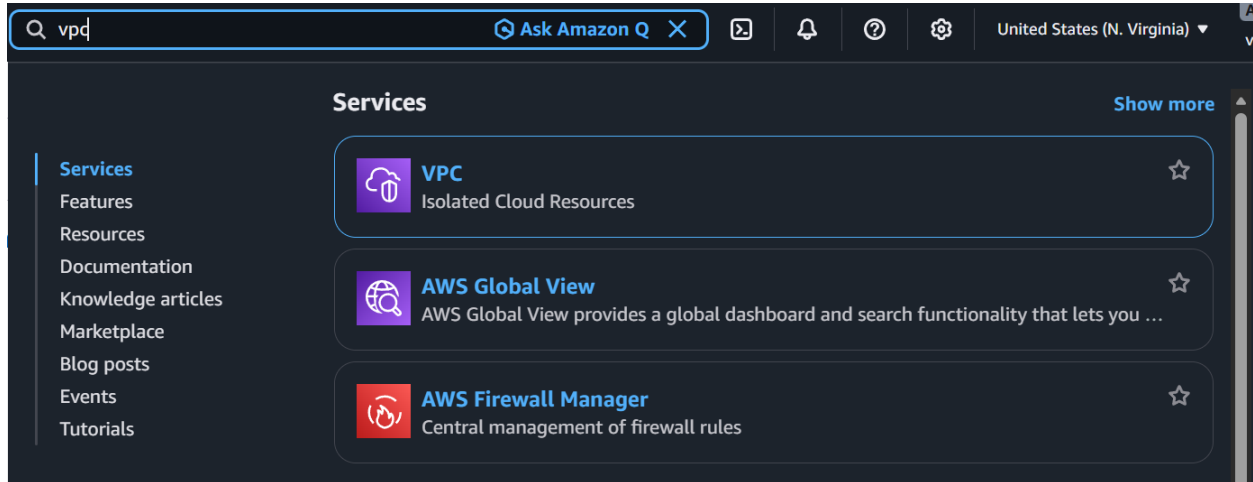
Lab Summary

This lab walks through the steps of setting up a database server within an AWS VPC or Virtual Private Cloud. In this lab a VPC is already set up and simply needs to be set up with a database server. This Database server is additionally replicated within another availability zone to provide better redundancy, resiliency, and availability.

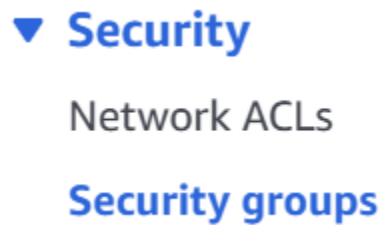
This lab uses a RDS or Relational Database server or structured data that is query able by SQL.

Lab Commands

Start and open the Lab. Once in the AWS Management Console search for the VPC service and select it.



Using the left navigation pane, navigate to security groups.

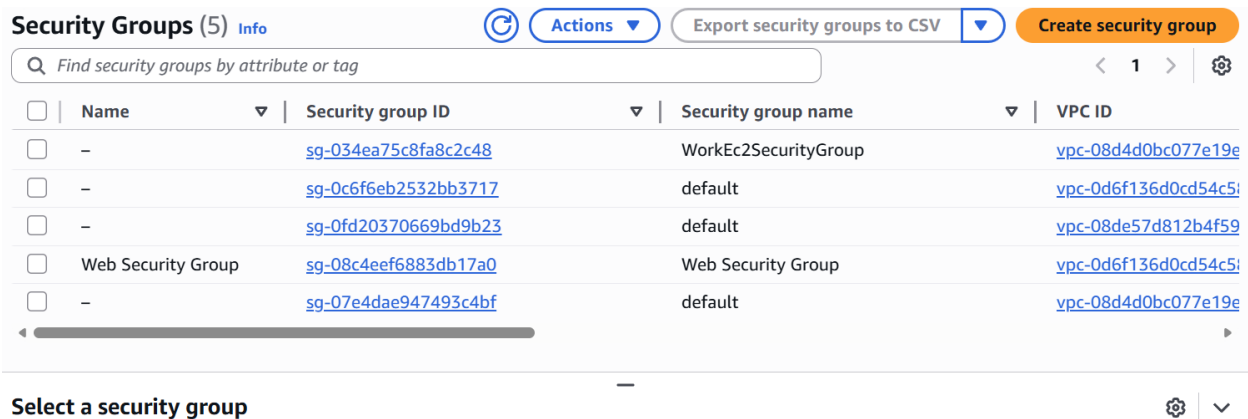


Choose to create security group and configure the following:

Name: DB Security Group

Description: Permit access from Web Security Group

VPC: Lab VPC



Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

Name cannot be edited after creation.

Description [Info](#)

VPC [Info](#)

Next, in the Inbound Rules pane choose to add a rule and configure it with the following:

Type: MySQL/Aurora

Source: type “sg” and select “Web Security Group”

Create the security group

Inbound rules [Info](#)

Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info
MySQL/Aurora	TCP	3306	Cu... sg-08c4eef6883db17a0	
Add rule				

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

Name cannot be edited after creation.

Description [Info](#)

VPC [Info](#)

Inbound rules [Info](#)

Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info
MySQL/Aurora	TCP	3306	Cu... sg-08c4eef6883db17a0	
Add rule				

Outbound rules [Info](#)

Type Info	Protocol Info	Port range Info	Destination Info	Description - optional Info
All traffic	All	All	Cu... 0.0.0.0/0	
Add rule				

Rules with destination of 0.0.0.0/0 or ::/0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

Add new tag

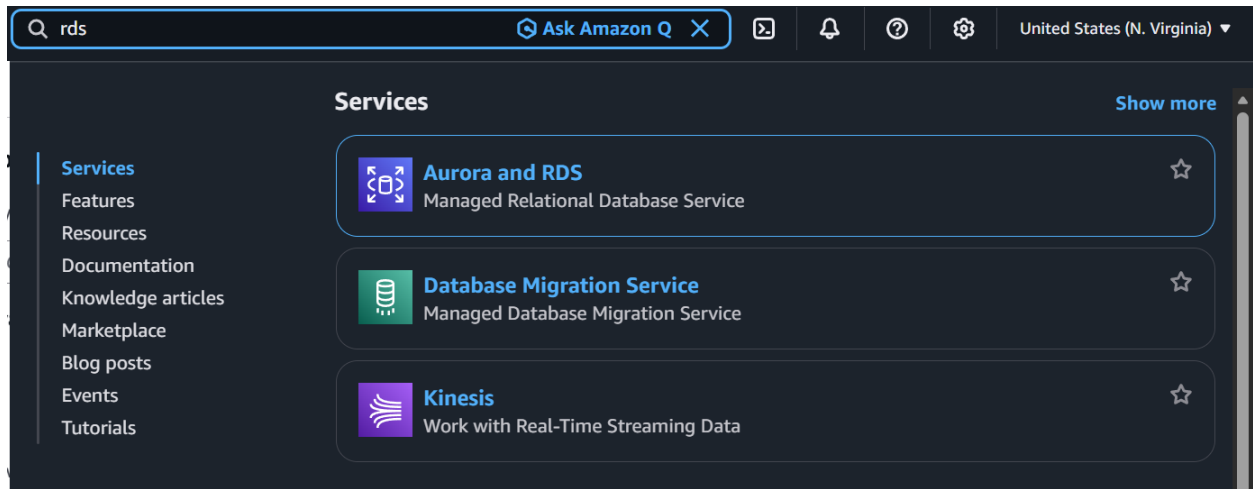
You can add up to 50 more tags

Cancel

Create security group

Next, create a DB subnet group used to tell RDS which subnets can be used for the database. Each DB subnet group requires two AZs.

In the console search for RDS and select it.



Navigate to subnet groups using the left navigation panel

Subnet groups

Parameter groups

Option groups

Custom engine versions

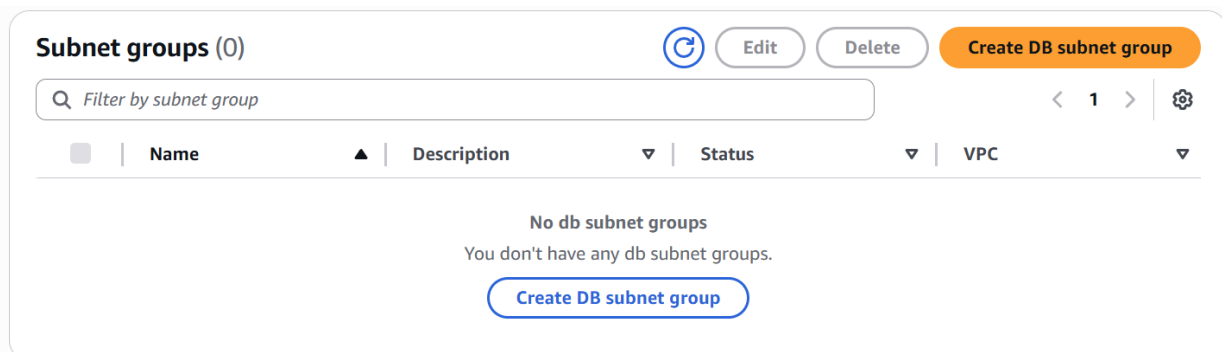
Zero-ETL integrations

Choose to create a subnet group with the following

Name: DB-Subnet-Group

Description: DB Subnet Group

VPC: Lab VPC



Scroll down to the add subnets section, click availability zones and select the first two zones: us-east-1a and us-east-1b. click on subnets and select the ranges 10.0.1.0/24 and 10.0.3.0/24.

Create DB subnet group

To create a new subnet group, give it a name and a description, and choose an existing VPC. You will then be able to add subnets related to that VPC.

Subnet group details

Name
You won't be able to modify the name after your subnet group has been created.

Must contain from 1 to 255 characters. Alphanumeric characters, spaces, hyphens, underscores, and periods are allowed.

Description

VPC
Choose a VPC identifier that corresponds to the subnets you want to use for your DB subnet group. You won't be able to choose a different VPC identifier after your subnet group has been created.

Lab VPC (vpc-0d6f136d0cd54c580)
4 Subnets, 2 Availability Zones

Add subnets

Availability Zones
Choose the Availability Zones that include the subnets you want to add.

Choose an availability zone

us-east-1a × us-east-1b ×

Subnets
Choose the subnets that you want to add. The list includes the subnets in the selected Availability Zones.

Select subnets

Private Subnet 1
Subnet ID: subnet-05be68afc1f5b3e82 CIDR: 10.0.1.0/24

Private Subnet 2
Subnet ID: subnet-0ace7373e0f08f288 CIDR: 10.0.3.0/24

For Multi-AZ DB clusters, you must select 3 subnets in 3 different Availability Zones.

Availability zone	Subnet name	Subnet ID	CIDR block
us-east-1a	Private Subnet 1	subnet-05be68afc1f5b3e82	10.0.1.0/24
us-east-1b	Private Subnet 2	subnet-0ace7373e0f08f288	10.0.3.0/24

Cancel Create

Subnets

Choose the subnets that you want to add. The list includes the subnets in the selected Availability Zones.

Select subnets

us-east-1b

☒ Private Subnet 2
Subnet ID: subnet-0ace7373e0f08f288 CIDR: 10.0.3.0/24

☐ Public Subnet 2
Subnet ID: subnet-031e8142001ae8745 CIDR: 10.0.2.0/24

us-east-1a

☐ Public Subnet 1
Subnet ID: subnet-0852497545b5c01a1 CIDR: 10.0.0.0/24

☒ Private Subnet 1
Subnet ID: subnet-05be68afc1f5b3e82 CIDR: 10.0.1.0/24

Next create a RDS MySQL database instance. In the left navigation pane choose Databases.

Aurora and RDS



Dashboard

Databases

Query editor

Performance insights

Snapshots

Exports in Amazon S3

Automated backups

Reserved instances

Proxies

Choose Create database and select MySQL under Engine Options. Under Templates choose dev/test. Under availability and durability choose Multi-AZ DB instance. Under setting configure:

DB instance identifier: lab-db

Master username: main

Master password: lab-password

Confirm password: lab-password

Databases (0)

☒ Group resources

Modify

Actions ▼

Create database ▼

< 1 >

DB identifier

▲

Status ▼

Role ▼

Engine ▼

Upgrade rollout order ▼

No resources

No resources to display

Create database ▼

Create database [Info](#)

Choose a database creation method

☒ **Full configuration**

You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ **Easy create**

Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Engine options

Engine type [Info](#)

☐ Aurora (MySQL Compatible)



☐ Aurora (PostgreSQL Compatible)



☒ **MySQL**



☐ PostgreSQL



☐ MariaDB



☐ Oracle



☐ Microsoft SQL Server



☐ IBM Db2



Templates

Choose a sample template to meet your use case.

☐ **Production**

Use defaults for high availability and fast, consistent performance.

☒ **Dev/Test**

This instance is intended for development use outside of a production environment.

☐ **Free tier**

Use RDS Free Tier to develop new applications, test existing applications, or gain hands-on experience with Amazon RDS. [Info](#)

Availability and durability

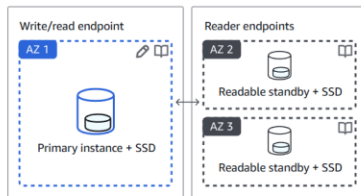
Deployment options [Info](#)

Choose the deployment option that provides the availability and durability needed for your use case. AWS is committed to a certain level of uptime depending on the deployment option you choose. Learn more in the [Amazon RDS service level agreement \(SLA\)](#).

☐ Multi-AZ DB cluster deployment (3 instances)

Creates a primary DB instance with two readable standbys in separate Availability Zones. This setup provides:

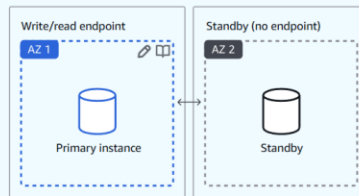
- 99.95% uptime
- Redundancy across Availability Zones
- Increased read capacity
- Reduced write latency



☒ **Multi-AZ DB instance deployment (2 instances)**

Creates a primary DB instance with a non-readable standby instance in a separate Availability Zone. This setup provides:

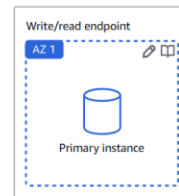
- 99.95% uptime
- Redundancy across Availability Zones



☐ Single-AZ DB instance deployment (1 instance)

Creates a single DB instance without standby instances. This setup provides:

- 99.5% uptime
- No data redundancy



Settings

DB instance identifier [Info](#)

Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

lab-db

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

Credentials Settings

Master username [Info](#)

Type a login ID for the master user of your DB instance.

main

1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management

You can use AWS Secrets Manager or manage your master user credentials.

☐

Managed in AWS Secrets Manager - *most secure*

RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒

Self managed

Create your own password or have RDS create a password that you manage.

☐

Auto generate password

Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

Password strength

Average

Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / ' * @

Confirm master password [Info](#)

▼ Additional credentials settings

Database authentication options [Info](#)

☒

Password authentication

Authenticates using database passwords.

☐

Password and IAM database authentication

Authenticates using the database password and user credentials through AWS IAM users and roles.

☐

Password and Kerberos authentication

Choose a directory in which you want to allow authorized users to authenticate with this DB instance using Kerberos Authentication.

Under DB instance class configure burstable classes with db.t3.micro

Instance configuration

The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class [Info](#)

▼ Hide filters

☒

Show instance classes that support Amazon RDS Optimized Writes [Info](#)

Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.

☒

Include previous generation classes

☐

Standard classes (includes m classes)

☐

Memory optimized classes (includes r and x classes)

☒

Burstable classes (includes t classes)

Instance type

db.t3.micro

2 vCPUs 1 GiB RAM EBS Bandwidth: Up to 2,085 Mbps Network: Up to 5 Gbps

Under storage:

Storage type: General Purpose (SSD)

Allocated storage: 20

Storage

Storage type [Info](#)
Provisioned IOPS SSD (io2) storage volumes are now available.

General Purpose SSD (gp2)
Baseline performance determined by volume size

Allocated storage [Info](#)
20 GiB
Allocated storage value must be 20 GiB to 6,144 GiB

Additional storage configuration

For high-throughput workloads, we recommend provisioning at least 100 GiB of General Purpose (SSD) storage. Lower values might result in higher latencies when the initial I/O credit balance is used up. [Learn more](#)

Under connectivity: Virtual Private Cloud: Lab VPC

Connectivity [Info](#)

Compute resource
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☒ **Don't connect to an EC2 compute resource**
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☐ **Connect to an EC2 compute resource**
Set up a connection to an EC2 compute resource for this database.

Virtual private cloud (VPC) [Info](#)
Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

Lab VPC (vpc-0d6f136d0cd54c580)
4 Subnets, 2 Availability Zones

After a database is created, you can't change its VPC. Only VPCs with a corresponding DB subnet group are listed.

Under existing VPC security groups: Choose DB Security Group Unselect default

DB subnet group [Info](#)
Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

db-subnet-group
2 Subnets, 2 Availability Zones

Public access [Info](#)

☐ **Yes**
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

☒ **No**
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) [Info](#)
Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☒ **Choose existing**
Choose existing VPC security groups

☐ **Create new**
Create new VPC security group

Existing VPC security groups
Choose one or more options

DB Security Group

Uncheck enhanced monitoring under monitoring > additional configuration

Monitoring [Info](#)

Choose monitoring tools for this database. Database Insights provides a combined view of Performance Insights and Enhanced Monitoring for your fleet of databases. **Database Insights** pricing is separate from RDS monthly estimates. See [Amazon CloudWatch pricing](#).

☐ Database Insights - Advanced

- Retains 15 months of performance history
- Fleet-level monitoring
- Integration with CloudWatch Application Signals

☒ Database Insights - Standard

▼ Additional monitoring settings

Enhanced Monitoring, CloudWatch Logs and DevOps Guru

Enhanced Monitoring

☐ Enable Enhanced monitoring

Enabling Enhanced Monitoring metrics are useful when you want to see how different processes or threads use the CPU.

Log exports

Select the log types to publish to Amazon CloudWatch Logs

- ☐ Audit log
- ☐ Error log
- ☐ General log
- ☐ iam-db-auth-error log
- ☐ Slow query log

IAM role

The following service-linked role is used for publishing logs to CloudWatch Logs.

RDS service-linked role

Configure the following under additional configuration:

Initial database name: lab

Uncheck enables automatic backups

Uncheck enable encryption

▼ Additional configuration

Database options, encryption turned off, backup turned off, backtrace turned off, maintenance, CloudWatch Logs, delete protection turned off.

Database options

Initial database name [Info](#)

lab

If you do not specify a database name, Amazon RDS does not create a database.

DB parameter group [Info](#)

default:mysql8.4

Option group [Info](#)

default:mysql-8-4

Backup

☐ Enable automated backup

Creates a point-in-time snapshot of your database

☐ Enable encryption

Choose to encrypt the given instance. Master key IDs and aliases appear in the list after they have been created using the AWS Key Management Service console. [Info](#)

Maintenance

Auto minor version upgrade [Info](#)

☒ Enable auto minor version upgrade

Enabling auto minor version upgrade will automatically upgrade your database minor version. For limitations and more details, see [Automatically upgrading the minor engine version documentation](#).

Maintenance window [Info](#)

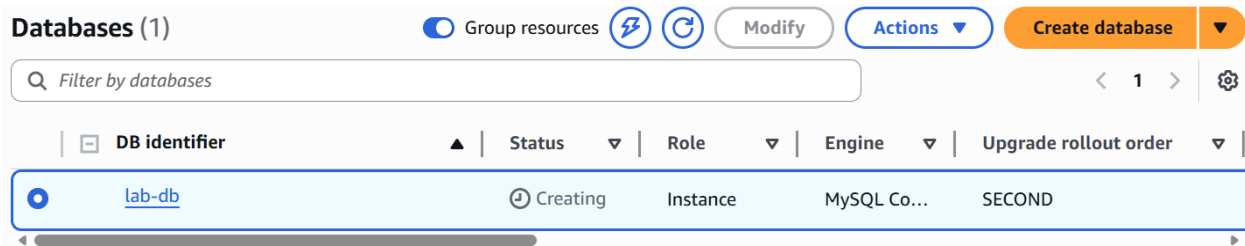
Select the period you want pending modifications or maintenance applied to the database by Amazon RDS.

- ☐ Choose a window
- ☒ No preference

☐ Enable deletion protection

Protects the database from being deleted accidentally. While this option is enabled, you can't delete the database.

Choose Create database, will not work if enhanced monitoring is enabled.



Wait for the server to be created by refreshing and checking the status.

Once the server is up, select it and scroll to the connectivity and security section and copy the endpoint and paste it into an editor.

Now Interact with the database. Find the IP address of the web server by going to the top of the Cloud Foundations lab and selecting “AWS Details” and paste it into a browser.

Once that window is open choose the RDS link at the top of the page and configure the following:

Endpoint: the DB endpoint stored in an editor

Database: lab

Username: main

Password: lab-password

Submit

aws Load Test RDS

Endpoint: lab-db.c2eyfusrifj.us-east-1.rds.amazonaws.com

Database: lab

Username: main

Password:

Submit

After a few seconds it will display an address book, test it by adding and removing contact. The data persists and is stored on the backup RDS server.

aws Load Test RDS

Address Book

Last name	First name	Phone	Email	Admin
Doe	Jane	010-110-1101	janed@someotheraddress.org	Add Contact Edit Remove
Johnson	Roberto	123-456-7890	roberto@someaddress.com	Edit Remove

You have finished the lab. Turn it in for full credit.

Total score	20/20
Task 1 - Security Group created	5/5
Task 2 - DB subnet group	5/5
Task 3 - DB created	5/5
Task 4 - App connected to DB	5/5

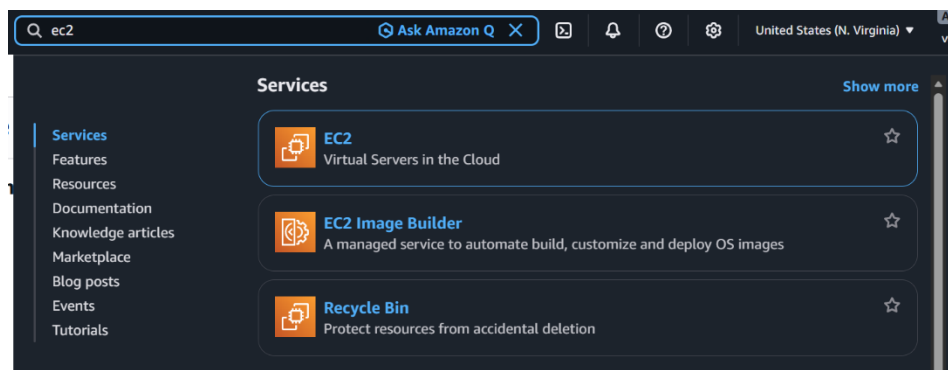
Lab 6 Walkthrough– Scale and Load Balance your Architecture

Lab Summary

This cloud foundations lab walks through the steps of how to set up EC2 autoscaling through autoscaling groups. This requires a load balancer as well configured to listen to traffic going to the EC2 instance web servers. This means that the load balancers are application load balancers balancing https traffic. This lab builds off the previous lab's architecture, including RDS databases inside private subnets. But only EC2 web server instances will be scaled.

Lab Commands

Start the Lab and navigate to the AWS management console. Then, navigate to the EC2 service. In the left navigation pane choose instance and wait until the Statues Checks for Web Server 1 displays 2/2 checks passed.



Instances (2) Info						
Find Instance by attribute or tag (case-sensitive)				All states ▾	< 1 > ⚙	
☐	Name ✎	Instance ID	Instance state ▾	Instance type ▾	Status check	Alarm status
☐	Bastion Host	i-0078eb7ac9f2eabc5	Running 🔍 🔍	t2.micro	⌚ Initializing	View alarms +
☐	Web Server 1	i-0d03927e9f9beeb16	Running 🔍 🔍	t2.micro	✅ 2/2 checks passed	View alarms +

Select Web Server 1 and in the actions menu choose Image and templates >
Create image and configure the following:

Image name WebServerAMI

Image description: Lab AMI for Web Server

Then create the image.

Instances (1/2) Info						
Find Instance by attribute or tag (case-sensitive)				All s	> ⚙	
☐	Name ✎	Instance ID	Instance state ▾	Instance type ▾		
☐	Bastion Host	i-0078eb7ac9f2eabc5	Running 🔍 🔍	t2.micro		
☒	Web Server 1	i-0d03927e9f9beeb16	Running 🔍 🔍	t2.micro		

Create image
Create template from instance
Launch more like this

Instance diagnostics
Instance settings ▶
Networking ▶
Security ▶
Image and templates ▶
Storage ▶
Monitor and troubleshoot ▶

Create image [Info](#)

An image (also referred to as an AMI) defines the programs and settings that are applied when you launch an EC2 instance. You can create an image from the configuration of an existing instance.

Image details

Instance ID

 i-0d03927e9f9beeb16 (Web Server 1)

Image name

Maximum 127 characters. Can't be modified after creation.

Image description - optional

Maximum 255 characters

Once the image is created, Create a load balancer. First, in the left navigation pane choose target groups and create a target group with the following:

Target Type: Instances

Target Group name: LabGroup

VPC: Lab VPC

Target groups [Info](#) | [What's new?](#)

[Actions](#)[Create target group](#)

< 1 >

☐	Name	ARN	Port	Protocol	Target type	L
--------------------------	------	-----	------	----------	-------------	---

No target groups

You don't have any target groups in us-east-1

[Create target group](#)

Settings - immutable

Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation."

Target type

Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

☒ Instances

Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.

Suitable for: [ALB](#) [NLB](#) [GWLB](#)

☐ IP addresses

Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.

Suitable for: [ALB](#) [NLB](#) [GWLB](#)

☐ Lambda function

Supports load balancing to a single Lambda function. ALB required as traffic source.

Suitable for: [ALB](#)

☐ Application Load Balancer

Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.

Suitable for: [NLB](#)

Target group name

Name must be unique per Region per AWS account.

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 8/32

Protocol

Protocol for communication between the load balancer and targets.

Port

Port number where targets receive traffic. Can be overridden for individual targets during registration.

1-65535

IP address type

Only targets with the indicated IP address type can be registered to this target group.

☒ IPv4

Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

☐ IPv6

Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC

Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

10.0.0.0/16[Create VPC](#)

Choose Next to continue, the register targets screen appears but there are not web application instances yet so this step can be skipped for now. Finish by clicking Create Target Group.

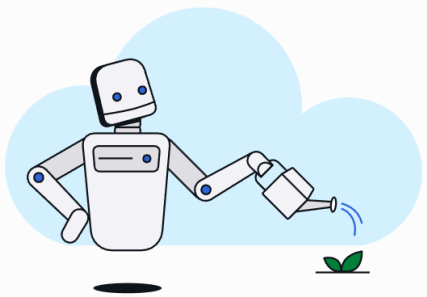
In the left navigation pane choose Load balancers and choose create load balancers. We are making an application load balancer.

Load balancers [What's new?](#)  [Actions](#) [Create load balancer](#)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

< 1 > 

	Name	State	Type	Scheme	IP address type	VPC ID
--	------	-------	------	--------	-----------------	--------

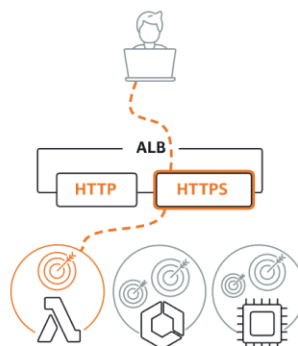


No load balancers
You don't have any load balancers in us-east-1

[Create load balancer](#)

Load balancer types

Application Load Balancer [Info](#)



Choose an Application Load Balancer when you need a flexible feature set for your applications with HTTP and HTTPS traffic. Operating at the request level, Application Load Balancers provide advanced routing and visibility features targeted at application architectures, including microservices and containers.

[Create](#)

Configure the following for the load balancer:

Name: LabELB

Scheme: Internet Facing

VPC: Lab VPC

Availability Zones and Subnets: Public Subnet 1 from First listed AZ

Public Subnet 2 from second displayed AZ

Security Groups: Web Security Group

Listener:

Protocol: HTTP

Port: 80

Routing Action Forward to Target Group

Target Group: LabGroup

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

LabELB

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme [Info](#)

Scheme can't be changed after the load balancer is created.

☒ Internet-facing

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ Internal

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type [Info](#)

Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

☒ IPv4

Includes only IPv4 addresses.

☐ Dualstack

Includes IPv4 and IPv6 addresses.

☐ Dualstack without public IPv4

Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with **internet-facing** load balancers only.

Network mapping [Info](#)

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC [Info](#)

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-0153d1063a730caf6 (Lab VPC)
10.0.0.0/16



[Create VPC](#)

IP pools [Info](#)

You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view **Pools** in the [Amazon VPC IP Address Manager console](#).

☐ Use IPAM pool for public IPv4 addresses

The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets [Info](#)

Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ us-east-1a (use1-az1)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-001f32f1d7963db55
IPv4 subnet CIDR: 10.0.0.0/24

Public Subnet 1

☒ us-east-1b (use1-az2)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-023d2915949a72f13
IPv4 subnet CIDR: 10.0.2.0/24

Public Subnet 2

Security groups [Info](#)

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups

Select up to 5 security groups

Web Security Group
sg-03b7ca66a4bdee3f8 VPC: vpc-0153d1063a730caf6

Listeners and routing [Info](#)

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

▼ Listener HTTP:80

Remove

Protocol

HTTP

Port

80

1-65535

Default action [Info](#)

The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action

☒ Forward to target groups

☐ Redirect to URL

☐ Return fixed response

Forward to target group [Info](#)

Choose a target group and specify routing weight or [create target group](#).

Target group

LabGroup

Target type: Instance, IPv4 | Target stickiness: Off

HTTP



Weight

1

0-999

Percent

100%

+ Add target group

You can add up to 4 more target groups.

Target group stickiness [Info](#)

Enables the load balancer to bind a user's session to a specific target group. To use stickiness the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.

☐ Turn on target group stickiness

Listener tags - optional

Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

Add listener tag

You can add up to 50 more tags.

Add listener

You can add up to 49 more listeners.

Now Create the Load balancer.

Next Create the Launch template for the autoscaling group. Navigate to Launch templates using the left navigation pane.

▼ Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager [New](#)

Click create Launch Template and configure the following:

Name: LabConfig

Auto Scaling Guidance: Opt in

AMI: Web Server AMI

Instance Type: t2.mirco

Key pair name: vockey

Firewall: Select existing security group

Security Groups: Web Security Groups

Now open the advanced details and enable detailed CloudWatch monitoring.

Create launch template

Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.

Launch template name and description

Launch template name - *required*

LabConfig

Must be unique to this account. Max 128 chars. No spaces or special characters like '&', '*', '@'.

Template version description

A prod webserver for MyApp

Max 255 chars

Auto Scaling guidance | [Info](#)

Select this if you intend to use this template with EC2 Auto Scaling

☒ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

► Template tags

► Source template

Launch template contents

Specify the details of your launch template below. Leaving a field blank will result in the field not being included in the launch template.

▼ Application and OS Images (Amazon Machine Image) - required [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

 Search our full catalog including 1000s of application and OS images

Recents

My AMIs

Quick Start

☒ Owned by me

☐ Shared with me



[Browse more AMIs](#)

Including AMIs from
AWS, Marketplace and
the Community

Amazon Machine Image (AMI)

WebServerAMI

ami-0aa3646e9ebb304dc

2026-02-09T21:15:41.000Z

Virtualization: hvm

ENA enabled: true

Root device type: ebs

Boot mode: uefi-preferred



Description

Lab AMI for Web Server

Architecture

x86_64

AMI ID

ami-0aa3646e9ebb304dc

▼ Instance type [Info](#) | [Get advice](#)

[Advanced](#)

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current generation: true

On-Demand Linux base pricing: 0.0116 USD per Hour

On-Demand Windows base pricing: 0.0162 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour

On-Demand SUSE base pricing: 0.0116 USD per Hour

On-Demand RHEL base pricing: 0.026 USD per Hour

☒ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name

vockey



[Create new key pair](#)

▼ Network settings [Info](#)

Subnet [Info](#)

Don't include in launch template ▼ [Create new subnet](#)

When you specify a subnet, a network interface is automatically added to your template.

Availability Zone [Info](#)

Don't include in launch template ▼ [Enable additional zones](#)

Not applicable for EC2 Auto Scaling

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Select existing security group ☐ Create security group

Security groups [Info](#)

Select security groups ▼

Web Security Group sg-03b7ca66a4bdee3f8 X
VPC: vpc-0153d1063a730caf6 [Compare security group rules](#)

► **Advanced network configuration**

Detailed CloudWatch monitoring [Info](#)

Enable ▼

Additional charges apply

Click the LabConfig template once the success dialog comes up.



Successfully created [LabConfig\(lt-01ed1dcd305cb44d8\)](#).

Then click actions menu and create an auto scaling group with the following:

Name: Lab Auto Scaling Group

Launch Template: LabConfig

VPC: Lab VPC

Availability Zones and Subnets: Private Subnet 1 and 2

Attach to an existing load balancer: Select the target LabGroup

Additional settings: Check group metrics collection

LabConfig (lt-01ed1dcd305cb44d8)

[Actions](#)

[Delete template](#)

Launch template details

Launch template ID

lt-01ed1dcd305cb44d8

Launch template name

LabConfig

Launch instance from template

Modify template (Create new version)

Delete template version

Set default version

Manage tags

Create Spot Fleet

Create Auto Scaling group

vs:sts::78349034014
-role/voclabs/user47
uki_Burleigh

[Details](#)

[Versions](#)

[Template tags](#)

Choose launch template or configuration [Info](#)

Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group. If you currently use launch configurations, you might consider migrating to launch templates.

Name

Auto Scaling group name

Enter a name to identify the group.

Lab Auto Scaling Group

Must be unique to this account in the current Region and no more than 255 characters.

Launch template [Info](#)

[Switch to launch configuration](#)

Launch template

Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

LabConfig

[Create a launch template](#)

Version

Default (1)

[Create a launch template version](#)

Description	Launch template	Instance type
-	LabConfig L	t2.micro
AMI ID	Security groups	Request Spot Instances
ami-0aa3646e9ebb304dc	-	No
Key pair name	Security group IDs	
vockey	sg-03b7ca66a4bdee3f8 L	

Additional details

Storage (volumes)	Date created
-	Mon Feb 09 2026 13:29:12 GMT-0800 (Pacific Standard Time)

Cancel

Next

Choose instance launch options [Info](#)

Choose the VPC network environment that your instances are launched into, and customize the instance types and purchase options.

Instance type requirements [Info](#)

[Override launch template](#)

You can keep the same instance attributes or instance type from your launch template, or you can choose to override the launch template by specifying different instance attributes or manually adding instance types.

Launch template	Version	Description
LabConfig L	Default	
lt-01ed1dcd305cb44d8		

Instance type

t2.micro

Network [Info](#)

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.

VPC

Choose the VPC that defines the virtual network for your Auto Scaling group.

vpc-0153d1063a730caf6 (Lab VPC)

[Create a VPC](#)

Availability Zones and subnets

Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets

use1-az1 (us-east-1a) | subnet-0052fd47f9704bc3b (Private Subnet 1)

use1-az2 (us-east-1b) | subnet-0d7638238701c756d (Private Subnet 2)

[Create a subnet](#)

Availability Zone distribution - new

Auto Scaling automatically balances instances across Availability Zones. If launch failures occur in a zone, select a strategy.

Balanced best effort

If launches fail in one Availability Zone, Auto Scaling will attempt to launch in another healthy Availability Zone.

Balanced only

If launches fail in one Availability Zone, Auto Scaling will continue to attempt to launch in the unhealthy Availability Zone to preserve balanced distribution.

Cancel

Skip to review

Previous

Next

Integrate with other services - *optional* [Info](#)

Use a load balancer to distribute network traffic across multiple servers. Enable service-to-service communications with VPC Lattice. Shift resources away from impaired Availability Zones with zonal shift. You can also customize health check replacements and monitoring.

Load balancing [Info](#)

Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

Select Load balancing options

☐

No load balancer

Traffic to your Auto Scaling group will not be fronted by a load balancer.

☒

Attach to an existing load balancer

Choose from your existing load balancers.

☐

Attach to a new load balancer

Quickly create a basic load balancer to attach to your Auto Scaling group.

Attach to an existing load balancer

Select the load balancers to attach

☒

Choose from your load balancer target groups

This option allows you to attach Application, Network, or Gateway Load Balancers.

☐

Choose from Classic Load Balancers

Existing load balancer target groups

Only instance target groups that belong to the same VPC as your Auto Scaling group are available for selection.

Select target groups



LabGroup | HTTP



Application Load Balancer: LabELB

Additional settings

Instance scale-in protection

If protect from scale in is enabled, newly launched instances will be protected from scale in by default.

☐

Enable instance scale-in protection

Monitoring [Info](#)

☒

Enable group metrics collection within CloudWatch

Default instance warmup [Info](#)

The amount of time that CloudWatch metrics for new instances do not contribute to the group's aggregated instance metrics, as their usage data is not reliable yet.

☐

Enable default instance warmup

Auto Scaling group deletion protection - *new* [Info](#)

Configure how this Auto Scaling group can be deleted.

☒

None (default)

The Auto Scaling group can be deleted.

☐

Prevent force deletion

The Auto Scaling group cannot be force deleted. To delete the group, all instances must be detached or terminated, and all warm pools must be deleted.

☐

Prevent all deletion

The Auto Scaling group cannot be deleted.

Next configure the minimum and maximum capacity for the group. This determines the minimum and maximum amount of EC2 instances.

Desired Capacity: 2

Minimum Capacity: 2

Maximum Capacity: 6

Configure group size and scaling - optional [Info](#)

Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.

Group size [Info](#)

Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

Desired capacity type

Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances) ▼

Desired capacity

Specify your group size.

2

Scaling [Info](#)

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity

2

Equal or less than desired capacity

Max desired capacity

6

Equal or greater than desired capacity

Automatic scaling - optional

Choose whether to use a target tracking policy [Info](#)

You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.



No scaling policies

Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.



Target tracking scaling policy

Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

Scaling policy name

LabScalingPolicy

Metric type [Info](#)

Monitored metric that determines if resource utilization is too low or high. If using EC2 metrics, consider enabling detailed monitoring for better scaling performance.

Average CPU utilization ▼

Target value

60

Instance warmup [Info](#)

300

seconds

☐ Disable scale in to create only a scale-out policy

Under scaling policies choose target tracking scaling policy and configure the following:

name: LabScalingPolicy

Metric type: Average CPU Utilization

Target Value: 60

Scaling [Info](#)

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity	Max desired capacity
2	6
Equal or less than desired capacity	Equal or greater than desired capacity

Automatic scaling - optional

Choose whether to use a target tracking policy | [Info](#)
You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

☐ **No scaling policies**

Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

☒ **Target tracking scaling policy**

Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

Scaling policy name**Metric type** | [Info](#)

Monitored metric that determines if resource utilization is too low or high. If using EC2 metrics, consider enabling detailed monitoring for better scaling performance.

Target value**Instance warmup** | [Info](#) seconds☐ Disable scale in to create only a scale-out policy

Now configure a tag to apply to any EC2 instance created with the autoscaling group:

Key: Name

Value: Lab Instance

Tags (1)

Key

Value - optional

Tag new instances

Name

Lab Instance

☒

Remove

Add tag

49 remaining

Cancel

Previous

Next

Review the details and create your autoscaling group.

Now to verify the load balancing look at the 2 new EC2 instances created through the autoscaling group in the instances page using the left navigation panel.

Instances (4) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Find Instance by attribute or tag (case-sensitive)

All states

< 1 >

⚙️

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☐	Lab Instance	i-0f517199b3d66502d	Running	t2.micro	Initializing	View alarms +
☐	Bastion Host	i-0078eb7ac9f2eabc5	Running	t2.micro	2/2 checks passed	View alarms +
☐	Web Server 1	i-0d03927e9f9beeb16	Running	t2.micro	2/2 checks passed	View alarms +
☐	Lab Instance	i-04acbc5d4767ac55	Running	t2.micro	Initializing	View alarms +

Select an instance

Now navigate to Target Groups, select LabGroup and choose the targets or details tab and notice how there are 2 targets.

Target groups (1/1) Info | What's new?

Actions

Create target group

Filter target groups

< 1 >

⚙️

☒	Name	ARN	Port	Protocol	Target type
☒	LabGroup	arn:aws:elasticloadbalancing:us-east-1:783490340141:targetgroup/LabGroup/f7fc7f6aca7bf09c	80	HTTP	Instance

Target group: LabGroup

Details

Targets

Monitoring

Health checks

Attributes

Tags

arn:aws:elasticloadbalancing:us-east-1:783490340141:targetgroup/LabGroup/f7fc7f6aca7bf09c

Target type

Instance

IP address type

IPv4

Protocol : Port

HTTP: 80

Load balancer

LabELB

Protocol version

HTTP1

VPC

vpc-0153d1063a730caf6

2

0

0

0

0

Total targets

Healthy

Unhealthy

Unused

Initial

Draining

0 Anomalous

Distribution of targets by Availability Zone (AZ)

Select values in this table to see corresponding filters applied to the Registered targets table below.

Now navigate to load balancers on the left navigation panel, select the LabELB load balancer and copy the DNS from the details pane.

Load balancers (1/1) [What's new?](#) [Actions](#) [Create load balancer](#)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers

Name	State	Type	Scheme	IP address type	VPC ID
LabELB	Active	application	Internet-facing	IPv4	vpc-0153d1

Load balancer: LabELB

[Details](#) [Listeners and rules](#) [Network mapping](#) [Resource map](#) [Security](#) [Monitoring](#) [Integrations](#)

Details

Load balancer type Application	Status Active	VPC vpc-0153d1063a730caf6	Load balancer IP address type IPv4
Scheme Internet-facing	Hosted zone Z35SXDOTRQ7X7K	Availability Zones subnet-023d2915949a72f13 us-east-1b (use1-az2) subnet-001f32f1d7963db55 us-east-1a (use1-az1)	Date created February 9, 2026, 13:24 (UTC-08:00)
Load balancer ARN arn:aws:elasticloadbalancing:us-east-1:783490340141:loadbalancer/app/LabELB/74cde4f49b5c156a		DNS name Info LabELB-581050631.us-east-1.elb.amazonaws.com (A Record)	

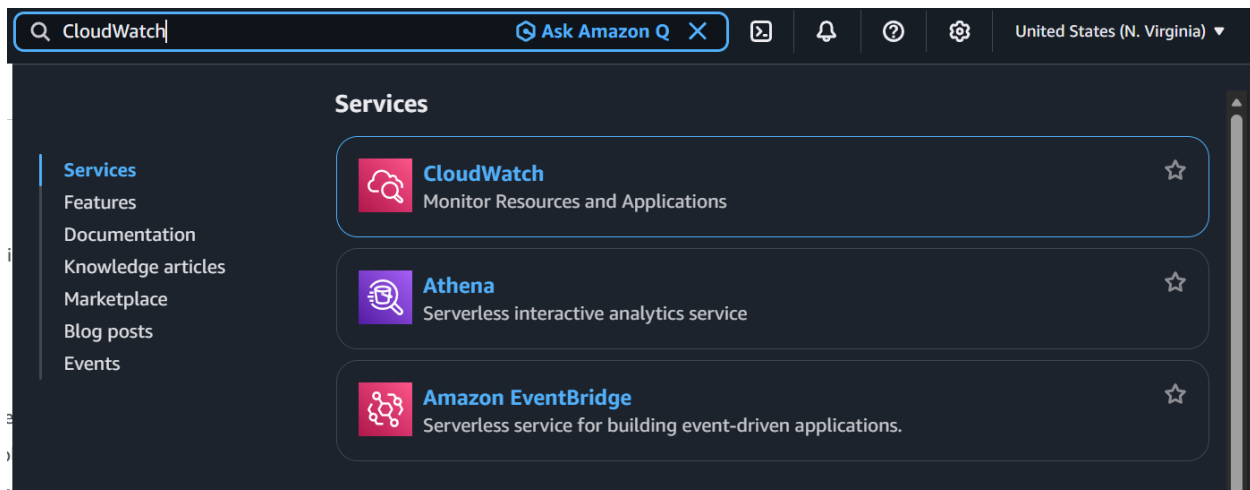
Open a new tab and paste the DNS address into a browser, if an application appears that means that the load balancer has sent your request to an EC2 instance and send the results back, indicating that the load balancing is working.

aws Load Test RDS

Meta-Data	Value
InstanceId	i-0f517199b3d66502d
Availability Zone	us-east-1b

Current CPU Load: 9%

Now test the actual autoscaling service. Don't close the newly made tab but navigate back to the console and to the CloudWatch service.



In the left navigation pane choose All alarms and two will be displayed. If they are not displayed follow these steps:

- Navigate back to EC2 service and go to autoscaling groups
- Select the Lab Auto Scaling Group > Automatic Scaling tab
- Select LabScalingPolicy > Actions > Edit > set Target Value to 50
- Choose Update and navigate back to CloudWatch

Among the alarms one should say AlarmHigh and the other AlarmLow. When the CPU utilization is high, AlarmHigh will change from OK to In Alarm, opposite for AlarmLow.

Alarms (2)

☐ Hide Auto Scaling alarms Clear selection Create composite alarm Actions Create alarm

Alarm state: Any Alarm type: Any

Actions status: Any < 1 > Settings

☐	Name	State	Actions	Last state update (UTC)	Cor
☐	TargetTracking-Lab Auto Scaling Group-AlarmLow-cd250f12-2c3d-47a3-b9d2-32f4060ee282	⚠ In alarm	✅ Actions enabled	2026-02-09 21:57:25	CPU mir
☐	TargetTracking-Lab Auto Scaling Group-AlarmHigh-8728031b-9552-458d-ae5d-a67004bf591a	ⓘ Insufficient data	✅ Actions enabled	2026-02-09 21:56:52	CPU mir

Now go back to the tab with the load balancer's DNS and click load test next to the AWS logo and you should see CPU load changing to 100% and new EC2 instances created when navigating back to EC2 > instances.

Generating CPU Load! (auto refresh in 5 seconds)

Current CPU Load: 100%

Instances (6) [Info](#)

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Find Instance by attribute or tag (case-sensitive)

All states

< 1 >

⚙

☐	Name ↗	Instance ID	Instance state ▼	Instance type ▼	Status check	Alarm status
☐	Lab Instance	i-0f517199b3d66502d	Running 🔍 🔍	t2.micro	2/2 checks passed	View alarms +
☐	Lab Instance	i-085af4bb0812420a9	Running 🔍 🔍	t2.micro	Initializing	View alarms +
☐	Bastion Host	i-0078eb7ac9f2eabc5	Running 🔍 🔍	t2.micro	2/2 checks passed	View alarms +
☐	Web Server 1	i-0d03927e9f9beeb16	Terminated 🔍 🔍	t2.micro	-	View alarms +
☐	Lab Instance	i-04acbc5d4767ac55	Running 🔍 🔍	t2.micro	2/2 checks passed	View alarms +
☐	Lab Instance	i-00d3e478c2e0c6741	Running 🔍 🔍	t2.micro	Initializing	View alarms +

Select an instance

Since Web Server 1 is no longer needed terminate it by selecting it terminating it under the instance state menu.

Turn the lab in for full credit.

Total score	35/35
Task 1 - AMI created	5/5
Task 2 - Load Balancer created	5/5
Task 3a - Launch Template created	5/5
Task 3b - Auto Scaling Group created	5/5
Task 4 - Load Balancer check	5/5
Task 5 - Auto Scaling check	5/5
Task 6 - Web Server 1	5/5

Problems

A problem during the lab I had was that sometimes I would accidentally create resources in different regions and are unable to connect them. This created a problem when trying to connect resources together within a VPC when one resource was in an entirely different region. To solve this, I would have to delete the resource and make sure I created it in N. Virginia so that they would be able to connect with each other and so that AWS would score my lab properly.

Another problem I had was during the EBS lab where I tried to attach a volume to the wrong device name which made it impossible to finish the lab, I eventually was able to solve this by going through all of the steps again to see what mistakes I made. I was able to resolve this and earn full points on the lab.

Conclusion

The series of AWS labs were very interesting as we were creating real applications that can be accessed globally instead. The series of AWS labs also helped me to develop an understanding of how to make the cloud secure and how AWS and other providers are able to let separate services communicate to each other in a secure way that can still be accessed globally. The labs were very fun as it was a new experience for me and as I said it was publicly accessible which made it a lot cooler and learning AWS and the Cloud much more enjoyable.